MECH 191 — Metallurgy

Week 6

Nonferrous Metals & Phase Diagrams

September 28, 2026 · 5:00 – 7:30 PM · Kai Santos

Class Agenda

01

Nonferrous Metals

Aluminum, copper, and titanium — what each family offers that steel cannot

02

Iron-Carbon Phase Diagram

Reading the map of steel microstructures — phases, regions, and key transformation points

03

Putting It Together

Technician's perspective on phase diagrams — and Unit 2 Quiz

Metallography Lab — Module 5

EXAMET-5-R-600-HD · 50–500×

Module 5 · Week 6

Nonferrous Metals & Phase Diagrams

What to do this week:

Observe twin density in brass vs. stainless steel. Look for fine Mg_2Si precipitates in Al at 500×.

Steps:

1. Observe — complete feature table
2. Sketch — label magnification & specimen
3. Answer — 4 short questions

Specimens to Check Out

- Specimen G — 6061-T6 aluminum
- Specimen K — 70/30 brass

How to check out the kit:

1. See instructor after class today
2. Sign out the specimen tray
3. Use the EXAMET-5 at your convenience
4. Return kit before leaving class
5. Submit completed module with your portfolio

Portfolio Handout — Module 5

MECH 191 — Course & Unit Roadmap

Unit 1 Materials Fundamentals	Unit 2 Steel & Alloys	Unit 3 Testing & Quality	Unit 4 Manufacturing	Unit 5 Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	**Wk 4** Sep 14 Carbon & Alloy Steels	**Wk 7** Oct 5 Mechanical Testing	**Wk 10** Oct 26 Casting	**Wk 14** Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	**Wk 5** Sep 21 Stainless & Cast Irons	**Wk 8** Oct 12 NDT Methods	**Wk 11** Nov 2 Forming	**Wk 15** Nov 30 Weld Metallurgy & Failure
	Wk 6 ← here Sep 28 Nonferrous & Phase Diagrams	**Wk 9** Oct 19 Quality Standards	**Wk 12** Nov 9 Machining	**Wk 16** Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 • FINAL EXAM

Week 6 — At a Glance

Topics

- Aluminum, copper, and titanium — why each family exists alongside steel
- Aluminum alloy designation (1xxx–7xxx) and heat-treatable vs. work-hardened series
- Brass and bronze — how copper alloys balance conductivity, strength, and corrosion resistance
- Titanium's alpha/beta microstructure and why Ti-6Al-4V dominates aerospace
- The iron-carbon phase diagram: ferrite, austenite, cementite, pearlite, and eutectoid
- CCT diagram — what actually forms during real cooling vs. equilibrium predictions
- Annealing, normalizing, quenching, tempering, and case hardening

Resources

Reference Material

Moniz Ch. 6 — Phase Diagrams

Kalpakjian Ch. 7
Nonferrous Metals and Alloys

Visualizers

Iron-Carbon Phase Diagram (Interactive)
Solidification & Dendrite Growth

Unit 2 Quiz

Covers Weeks 4–6

Core Questions

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

1. What are the three major nonferrous metal families and what unique advantage does each offer that steel cannot match?
2. What does the iron-carbon phase diagram tell you — and what critical information does it NOT give you?
3. What is the difference between an iron-carbon phase diagram and a CCT diagram, and when do you need each one?
4. How does cooling rate determine the microstructure — and therefore the mechanical properties — of steel?
5. Quenched martensite is extremely hard but is almost never used in that condition. What does tempering do, and what trade-off does it involve?

Nonferrous Metals — Why They Exist Alongside Steel

Aluminum

1/3 the density of steel

- ~2.7 g/cm^3 — dominant in aerospace, automotive, and structural applications where weight matters
- Naturally corrosion-resistant — stable Al_2O_3 passive layer (similar to stainless)
- 1xxx–7xxx designation system; heat-treatable (2xxx, 6xxx, 7xxx) and work-hardened (1xxx, 3xxx, 5xxx) series
- Melts at ~660°C — much lower than steel; weldable but requires different process (GTAW/GMAW)

Copper

Best electrical conductivity

- Standard for electrical wiring, busbars, and heat exchangers — no common structural metal matches it
- Brass (Cu-Zn): good strength, machinability, corrosion resistance — fittings, valves, decorative
- Bronze (Cu-Sn or Cu-Al/Si): higher strength and wear resistance — bearings, bushings, marine hardware
- High thermal conductivity demands greater heat input when welding compared to steel

Titanium

Best strength-to-weight ratio

- ~60% the density of steel, with comparable or superior strength — used in aerospace, medical implants
- Corrosion resistance exceeds even the best stainless in most environments
- Alpha (HCP), beta (BCC), and alpha-beta alloys — Ti-6Al-4V (Grade 5) is the workhorse alloy
- Extremely reactive at elevated temps — must weld in inert shielding or vacuum to prevent embrittlement

Aluminum Alloys — Designation and Selection

4-Digit Designation System

Series	Alloying Element	Strengthening	Common Use
1xxx	Pure Al (≥99%)	Work-hardened only	Electrical conductors, foil
2xxx	Cu alloying	Heat-treatable	Aerospace structures (7075 rival)
3xxx	Mn alloying	Work-hardened only	Beverage cans, roofing
5xxx	Mg alloying	Work-hardened only	Marine, structural, pressure vessels
6xxx	Mg + Si alloying	Heat-treatable	Most widely used — extrusions, frames
7xxx	Zn alloying	Heat-treatable	Highest-strength Al — aircraft skins

Technician Notes

Melting point

~660°C — far lower than steel. High heat input in welding can cause burn-through or hot cracking without proper technique.

Weld process

GTAW (TIG) or GMAW (MIG) with inert shielding. AC polarity for GTAW to break up the tenacious Al_2O_3 surface oxide.

Heat treatment

T6 temper (solution treat + age) is the most common condition for 6061 and 7075 — provides near-peak strength.

Galvanic risk

Aluminum is anodic relative to steel and copper. Avoid direct contact in wet environments; use isolation gaskets.

Copper & Titanium — Key Properties and Alloy Families

Copper Alloys

Best electrical & thermal conductivity of structural metals

- Pure copper (C10000–C18000): highest conductivity — standard for wiring, busbars, heat exchangers
- Brass (Cu-Zn): improved strength and machinability over pure Cu — fittings, valves, fasteners, decorative parts
- Bronze (Cu-Sn): better wear resistance than brass — bearings, bushings, gear blanks, marine hardware
- Cu-Al / Cu-Si / Cu-Be bronzes: specialist alloys for springs, corrosion-resistant bolts, non-sparking tools
- Welding: high thermal conductivity demands greater heat input than steel; most grades weld with GTAW or GMAW; oxygen-free grades preferred to prevent porosity

Titanium Alloys

Best strength-to-weight ratio · exceptional corrosion resistance

- Density ~4.5 g/cm^3 (60% of steel) with comparable or superior strength — indispensable in aerospace
- Alpha alloys (HCP): best creep resistance at high temperature; weldable; CP titanium used in chemical processing
- Beta alloys (BCC): highest strength, heat-treatable; used for fasteners and springs
- Alpha-beta: Ti-6Al-4V (Grade 5) — accounts for ~50% of all titanium used; dominant in aircraft frames, jet engine fan blades, medical implants
- Welding: extremely reactive above 600°C — must use inert gas trailing shield or weld in a purge box/vacuum; any oxygen or nitrogen pickup causes brittle embrittlement

The Iron-Carbon Phase Diagram

What It Tells You

- Maps the stable phases in the iron-carbon system from pure iron (0% C) to iron carbide (6.67% C = Fe_3C)
- Horizontal axis: carbon content (%). Vertical axis: temperature (°C). Any point = equilibrium microstructure at that T and composition
- Crossing a phase boundary during heating or cooling triggers a phase transformation
- The diagram assumes equilibrium — it shows what should form if you hold temperature long enough for the microstructure to fully stabilize
- It does NOT account for cooling rate — that is what the CCT diagram is for

Key Regions & Points

Ferrite (α):
BCC, max 0.022% C at 727°C — soft and ductile; stable phase in low-carbon steels at room temperature

Austenite (γ):
FCC, dissolves up to 2.14% C at 1147°C — the phase steel must be in for most heat treatment to work

Cementite (Fe_3C):
6.67% C — extremely hard, brittle iron carbide; forms when carbon exceeds solubility limit

Pearlite:
Not a phase — a microstructural constituent: lamellar ferrite + cementite formed at eutectoid (0.77% C, 727°C)

Eutectoid point:
0.77% C at 727°C — austenite transforms simultaneously to ferrite + cementite; boundary between hypo- and hypereutectoid

Steel vs. Cast Iron:
2.14% C = boundary — above this, carbon can no longer fully dissolve in austenite

Reading the Phase Diagram — Steel Classification

Hypoeutectoid Steel

< 0.77% C

- On slow cooling from austenite, proeutectoid ferrite forms first at grain boundaries
- Remaining austenite then transforms to pearlite at 727°C
- Room-temperature microstructure: ferrite + pearlite
- More carbon → more pearlite → higher strength, lower ductility
- Examples: 1020 (0.20% C), 1040 (0.40% C) — most structural steels fall here

Eutectoid Steel

= 0.77% C

- Austenite transforms entirely to pearlite at exactly 727°C — no proeutectoid phase
- Finest microstructure achievable by slow cooling alone
- Maximum hardness from slow cooling, but martensite gives far higher hardness when quenched
- Example: 1080 (0.80% C) — wire rope, rail steel

Hypereutectoid Steel

> 0.77% C

- On slow cooling, proeutectoid cementite forms at prior austenite grain boundaries before pearlite
- Cementite network can cause brittleness — often spheroidize annealed to break up the network
- Room-temperature microstructure: cementite + pearlite
- Examples: 1095 (0.95% C), 52100 bearing steel (1.0% C) — tools, springs, bearings

The Technician's Perspective — Putting It Together

Material Selection	• Use aluminum (6061-T6) where weight matters and strength is moderate — most structural fabrication, frames, enclosures • Specify copper alloy based on function: pure Cu for conductivity; brass for fittings/machinability; bronze for bearings/wear • Titanium when steel is too heavy AND stainless isn't corrosion-resistant enough — budget for specialized welding procedures
Reading a WPS / Material Cert	• Phase diagram knowledge lets you interpret weld heat-affected zone behavior — why HAZ can soften (Al) or harden (medium-C steel) • CCT diagram reasoning: if you see 'preheat to 200°C,' that's slowing the cooling rate to avoid martensite in the HAZ • Temper color on stainless or Ti welds tells you the surface temperature reached — purple = ~300°C, straw = ~200°C
Heat Treatment Specifications	• 'Normalize and temper' on a drawing means two separate heating cycles — not one • Tempering temperature is critical: 150°C gives a different hardness than 450°C; verify with hardness test after • Case depth on carburized parts is specified on drawing — confirm by sectioning a coupon and etching with nital

Core Questions — Answers

1

What are the three major nonferrous families and what unique advantage does each offer?

Aluminum (one-third the density of steel — weight-critical apps), Copper (best electrical/thermal conductivity — wiring & heat exchangers), Titanium (best strength-to-weight ratio + exceptional corrosion resistance — aerospace & medical).

2

What does the iron-carbon phase diagram tell you — and what does it NOT tell you?

It maps which phases are stable at equilibrium for any temperature and carbon %. It does NOT show what forms during real industrial cooling — that requires the CCT diagram, which accounts for cooling rate.

3

What is the difference between the iron-carbon phase diagram and a CCT diagram?

Phase diagram = thermodynamic equilibrium — what forms if you hold temperature long enough. CCT = continuous cooling — what actually forms at different cooling rates. CCT is steel-specific; phase diagram covers the whole Fe-C system.

4

How does cooling rate determine the microstructure — and properties — of steel?

Slow → carbon diffuses fully → soft pearlite. Intermediate → bainite (stronger, tougher). Fast quench → carbon trapped → hard brittle martensite. CCT diagram shows exactly which cooling curve produces which result.

5

What does tempering do to as-quenched martensite, and what trade-off does it involve?

Tempering reheats martensite below the austenite region, letting some carbon diffuse and relieving internal stress. Trade-off: higher tempering temperature = more toughness but lower hardness. Required because as-quenched martensite is too brittle for service.

Key Takeaways — Week 6

01 The three major nonferrous families solve problems steel cannot: aluminum for weight, copper for conductivity, and titanium for corrosion resistance combined with high strength-to-weight. Knowing when to reach for a nonferrous alloy — and which one — is as important as knowing the steel families.

02 The iron-carbon phase diagram and the CCT diagram work together — the phase diagram shows what is thermodynamically possible at equilibrium; the CCT diagram shows what actually forms under real cooling conditions. You need both to reason about heat treatment.

03 Cooling rate is the most powerful processing variable in steel heat treatment: slow cooling gives soft pearlite, fast quenching gives hard martensite, and tempering after quenching trades some hardness back for toughness. This gives a technician the ability to reason about why a heat-treated part performs the way it does — and what went wrong when it doesn't.

Resources & What's Next

Reference Material

- Moniz 5th Edition — Ch. 6: Phase Diagrams
- Kalpakjian 6th Edition — Ch. 7: Nonferrous Metals and Alloys

Visualizers

- Iron-Carbon Phase Diagram (Interactive)
- Solidification & Dendrite Growth (Interactive)

Coming Up — Week 7

Mechanical Testing

- Tensile testing and the stress-strain curve — yield strength, UTS, elongation
- Hardness testing: Rockwell, Brinell, Vickers — when to use each
- Charpy impact testing — measuring toughness and ductile-to-brittle transition
- How testing results connect back to the microstructures we mapped this week

Unit 2 Quiz covers Weeks 4 – 6 · Review the Fe-C diagram and CCT microstructures